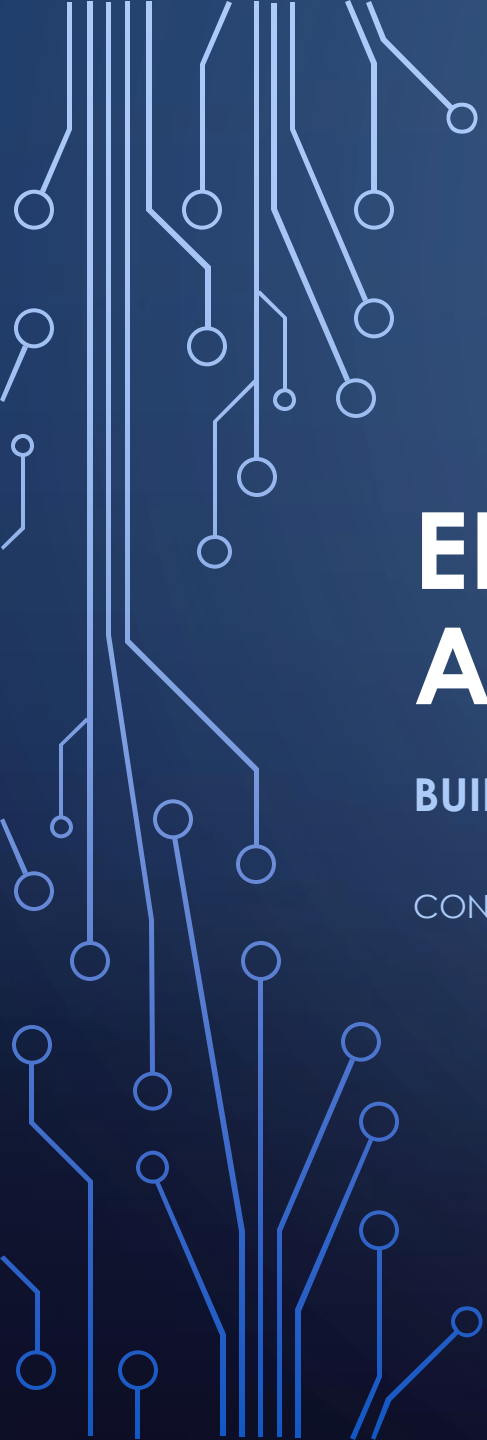




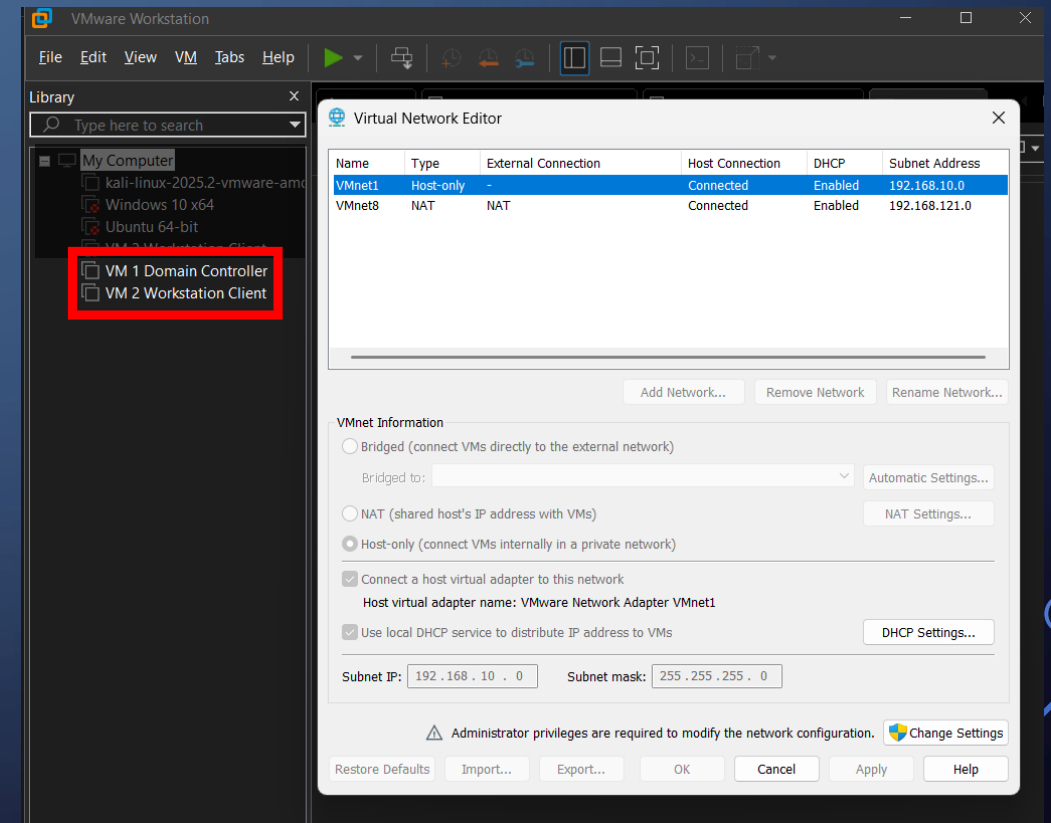
ENTERPRISE SYSTEMS INFRASTRUCTURE & AUTOMATION LAB

BUILDING A SECURE, AUTOMATED ACTIVE DIRECTORY CORE ENVIRONMENT

CONTEXT: DESIGNED TO SHOWCASE TIER 1-2 ENTERPRISE SYSTEMS AND INFRASTRUCTURE COMPETENCIES

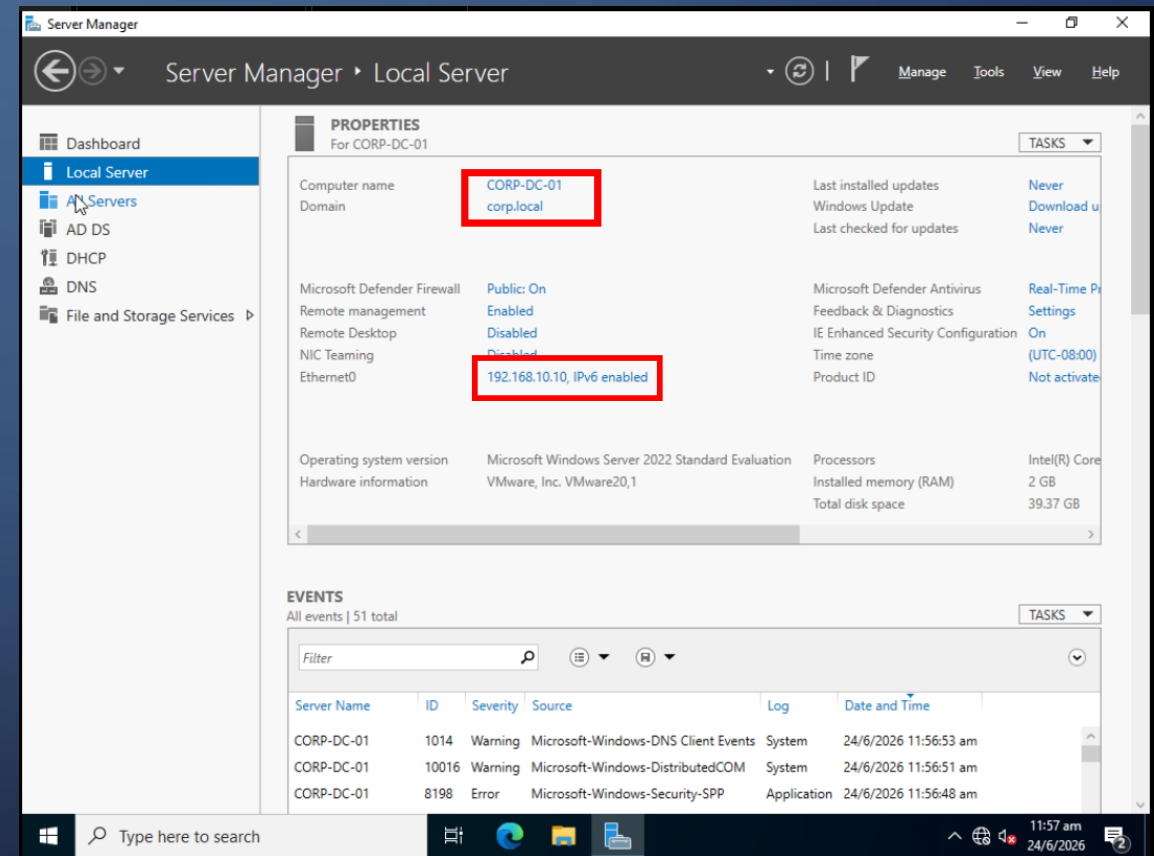
PROJECT ARCHITECTURE & SANDBOX DESIGN

- Hypervisor Isolation: VMware Workstation host-only private network (192.168.10.0/24)
- Risk Mitigation: Sandbox guarantees zero packet leakage/DHCP conflicts
- Node Distribution: Two enterprise VMs – Windows Server 2022 Standard & Windows 10 Enterprise



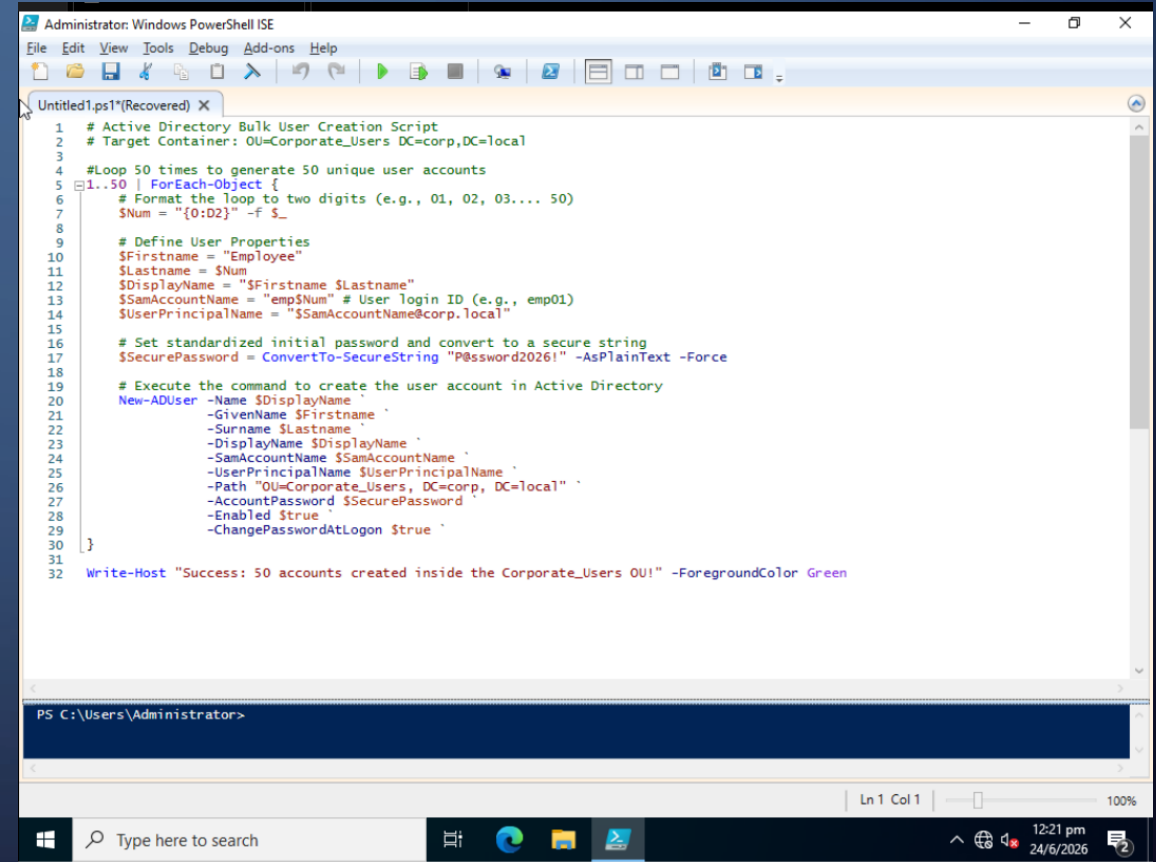
CORE INFRASTRUCTURE & STATIC NETWORKING

- Predictable Routing: Static IP (192.168.10.10) for Windows Server
- DNS Authority: Local loopback pointer (127.0.0.1) for directory lookups
- Forest Initialization: Promoted into root directory forest (corp.local)



ACTIVE DIRECTORY STRUCTURE & POWERSHELL AUTOMATION

- Logical Organization: Custom OU "Corporate_Users"
- Efficiency Focus: Automated account creation via PowerShell ISE
- Scalability: Script loop generated 50 accounts with password complexity in seconds



The screenshot displays the Windows PowerShell ISE interface. The main editor window shows a PowerShell script titled 'Untitled1.ps1 (Recovered)'. The script is designed to create 50 user accounts in an Active Directory container named 'OU=Corporate_Users, DC=corp, DC=local'. It uses a loop to generate unique usernames and passwords, and then executes the 'New-ADUser' command for each user. The script includes comments explaining each step, such as defining user properties, setting a secure password, and executing the 'New-ADUser' command. The console window at the bottom shows the command prompt 'PS C:\Users\Administrator>'.

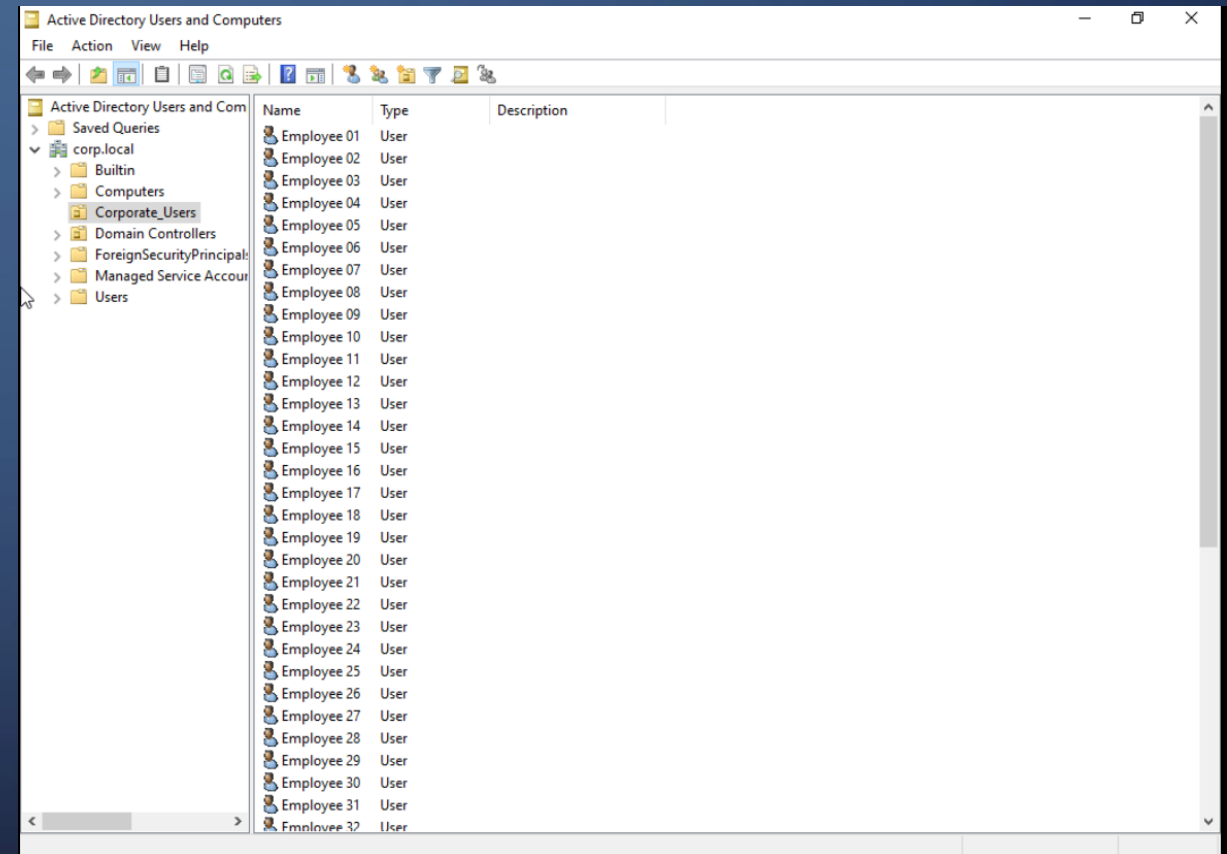
```
1 # Active Directory Bulk User Creation Script
2 # Target Container: OU=Corporate_Users DC=corp,DC=local
3
4 #Loop 50 times to generate 50 unique user accounts
5 1..50 | ForEach-Object {
6     # Format the loop to two digits (e.g., 01, 02, 03... 50)
7     $Num = "{0:D2}" -f $_
8
9     # Define User Properties
10    $Firstname = "Employee"
11    $Lastname = $Num
12    $DisplayName = "$Firstname $Lastname"
13    $SamAccountName = "emp$Num" # User login ID (e.g., emp01)
14    $UserPrincipalName = "$SamAccountName@corp.local"
15
16    # Set standardized initial password and convert to a secure string
17    $SecurePassword = ConvertTo-SecureString "P@ssword2026!" -AsPlainText -Force
18
19    # Execute the command to create the user account in Active Directory
20    New-ADUser -Name $DisplayName `
21        -GivenName $Firstname `
22        -Surname $Lastname `
23        -DisplayName $DisplayName `
24        -SamAccountName $SamAccountName `
25        -UserPrincipalName $UserPrincipalName `
26        -Path "OU=Corporate_Users, DC=corp, DC=local" `
27        -AccountPassword $SecurePassword `
28        -Enabled $true `
29        -ChangePasswordAtLogon $true `
30
31 }
32 Write-Host "Success: 50 accounts created inside the Corporate_Users OU!" -ForegroundColor Green
```

PS C:\Users\Administrator>

ACTIVE DIRECTORY STRUCTURE & POWERSHELL AUTOMATION

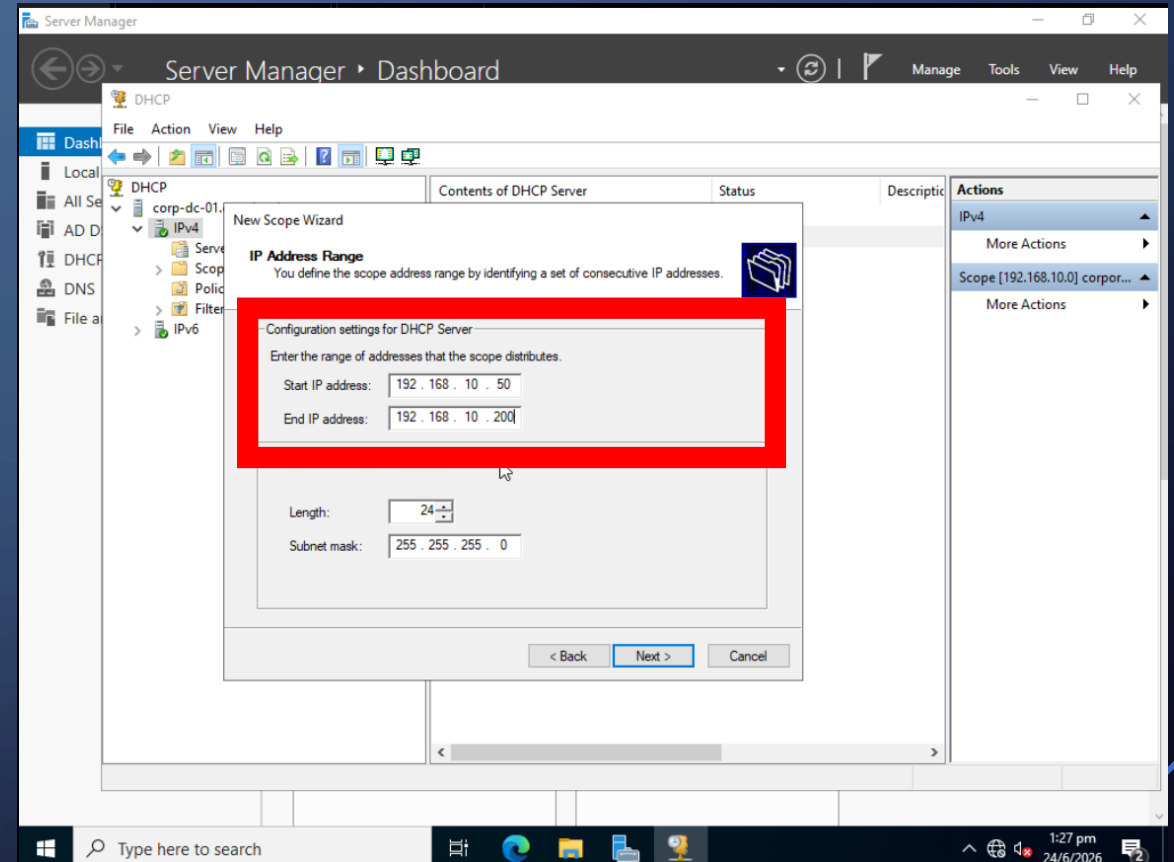
- CONTINUE

- Logical Organization: Custom OU "Corporate_Users"
- Efficiency Focus: Automated account creation via PowerShell ISE
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CORE NETWORK SERVICES (DHCP) & WORKSPACE SYNC

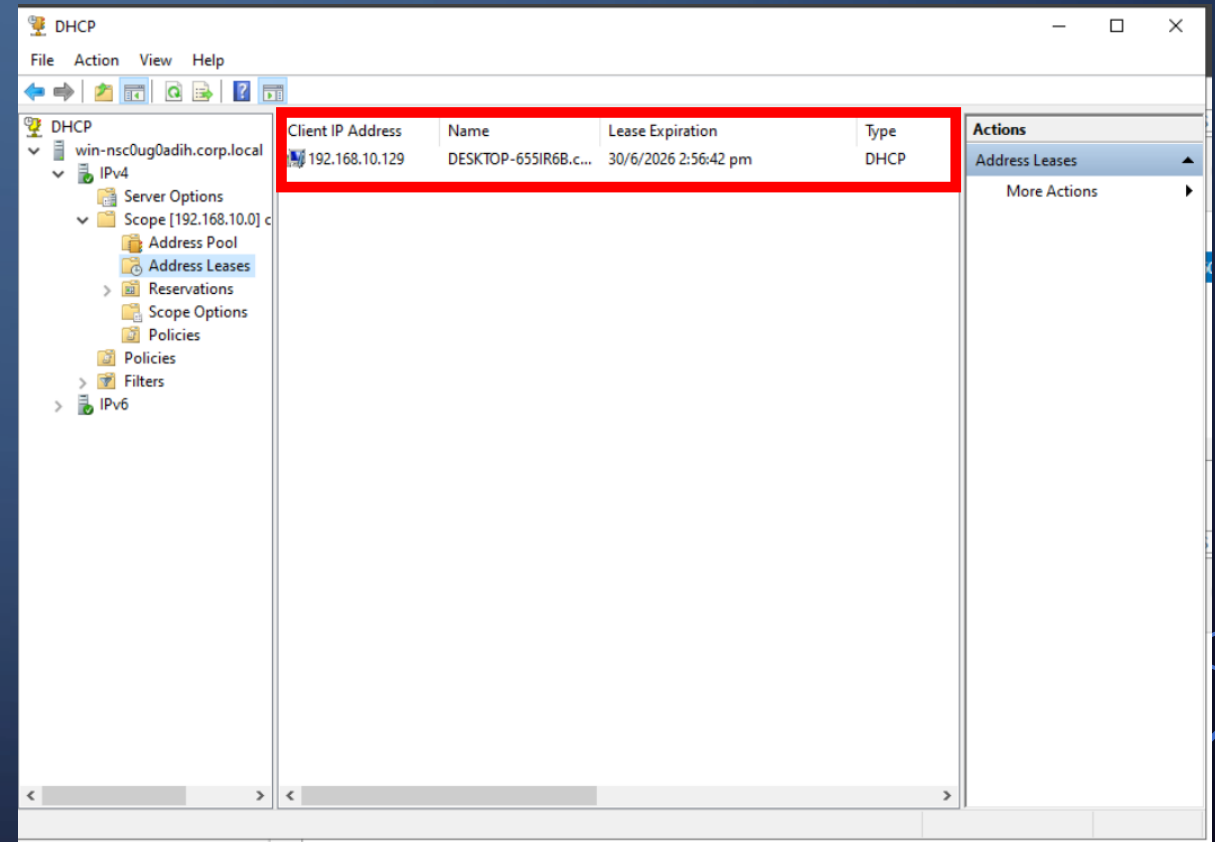
- Network Provisioning: DHCP role with dynamic range (.50-.200)
- Dynamic Client Lease: Windows 10 node captured lease at 192.168.10.129
- Centralized Management: Logs validated in Windows Server database console



CORE NETWORK SERVICES (DHCP) & WORKSPACE SYNC

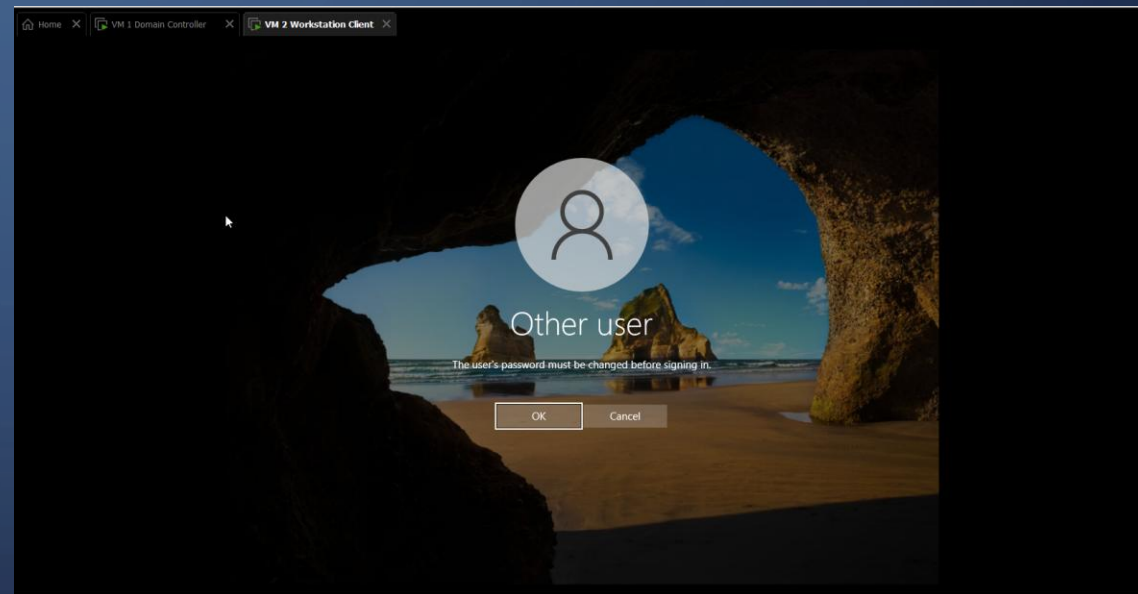
– CONTINUE

- Network Provisioning: DHCP role with dynamic range (.50–.200)
- Dynamic Client Lease: Windows 10 node captured lease at 192.168.10.129
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ENDPOINT VALIDATION & PROJECT RETROSPECTIVE

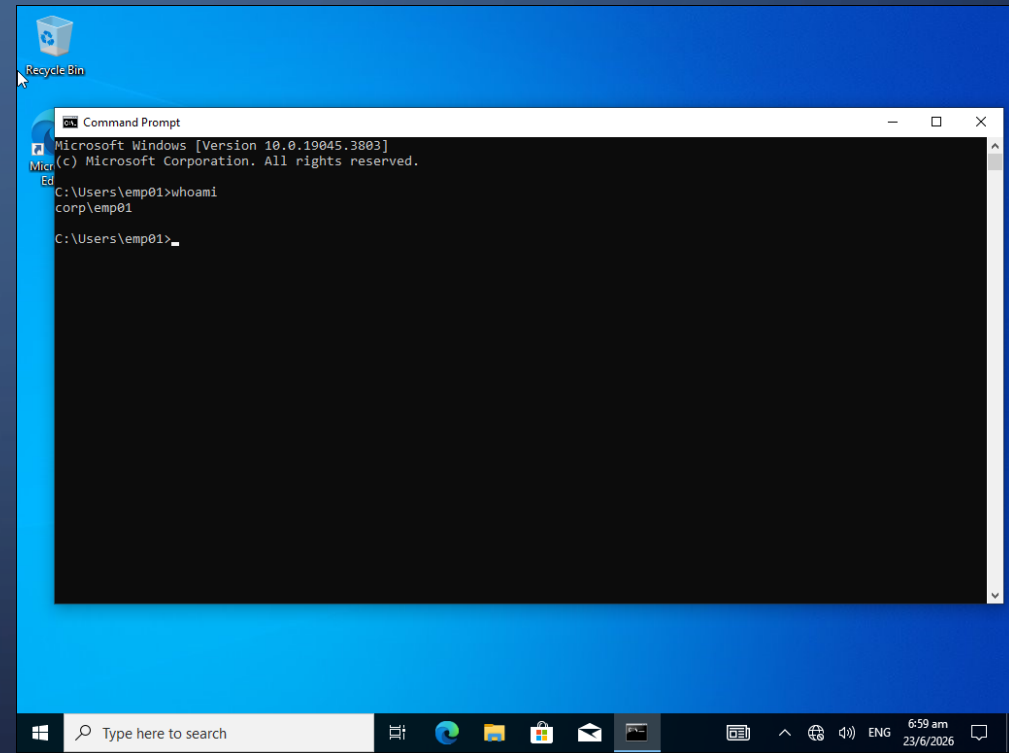
- Explicit Path Authentication: Domain routing (corp.local\Administrator)
- Synchronization Check: Automated account login (emp01), password reset validation
- Core Competencies Proven: Hypervisor Networking, Directory Lifecycle Management, Scripting/SysAdmin Automation, Enterprise Troubleshooting

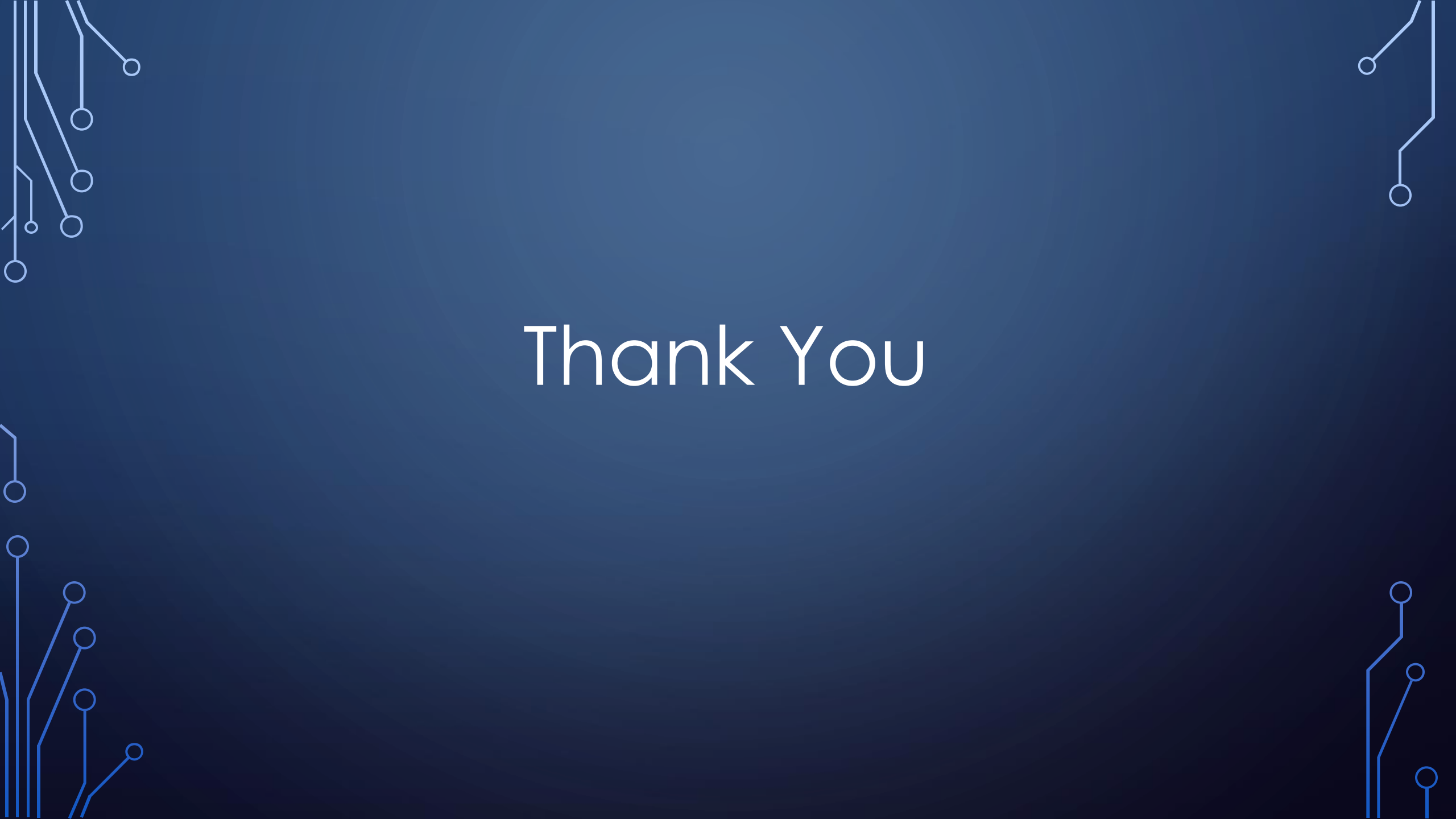


ENDPOINT VALIDATION & PROJECT RETROSPECTIVE

- CONTINUE

- Explicit Path Authentication: Domain routing (corp.local\Administrator)
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- Core Competencies Proven: Hypervisor Networking, Directory Lifecycle Management, Scripting/SysAdmin Automation, Enterprise Troubleshooting



The background is a dark blue gradient. In the corners, there are white line-art illustrations of circuit boards or neural networks. These consist of straight lines of varying lengths and small circles at the end of the lines, suggesting nodes or connections. The lines are more densely packed in the top-left and bottom-left corners, and more sparse in the top-right and bottom-right corners.

Thank You